Hey Vayro: from SIH prototype to product

Commercialization brief · Team DATA_Edgepulse · SIH 2026, PS 26172 (ISRO): Low Latency and Efficient Voice Activator for Edge Devices

If this goes beyond SIH, what can it become as a real product? This brief answers that in two pages.

TODAY · BUILT

- Working, physically tested prototype on real hardware
- Within both PS limits: 245.0 KiB RAM, 9.32% idle CPU
- Open source, with a public 104-page technical report

NEXT · PLANNED

- Fix false wake-ups from TV and conversation
- More voices and homes; secure, easy set-up
- Connect commands to real devices; pilot sites

EVENTUALLY · OPPORTUNITY

- Custom wake words for Indian device makers
- On-premise voice kits for labs, campuses and homes
- Support and update plans for installed devices

1 Product

A low-cost **voice activator** that listens for its own name on a small chip and sends audio only after it hears it. The chip decides *when* to listen; a speech server that the owner controls decides *what was said*. While idle it sends no audio, only a 72-byte clock-sync message every 10 seconds. What we would sell is that system packaged three ways: a **wake-word module** for products, a **custom wake-word service** ("Hey <your brand>"), and an **on-premise kit** (edge devices plus a local server).

1. You say "Hey Vayro, turn on the light" → **2. The chip** recognises its name, on the device, in about 0.1 s → **3. Only then** a few seconds of audio go to *your own* server → **4. The server** returns "turn on the light" as text, ready to act on

2 Target customers (first, not everyone)

Segment	Their problem	What we offer	Who pays
Indian device makers and IoT start-ups	Want their own wake word, but vendor routes need a licence or 20,000+ recordings from 500+ people plus a fee	Custom wake word, firmware and a measured test report	The company (B2B)
Labs, campuses, offices with their own network	Want hands-free voice without sending room audio outside the building	On-premise kit and installation	The institution (B2B)
Privacy-minded homes (via installers and open-source home hubs)	Want voice control without a vendor cloud	Free open firmware; ready-made kits	Homeowner or installer

Not our first market: factory floors, hospital nurse-call and hotel rooms. They need noise and speaker robustness we have not shown yet.

3 Value proposition: why choose this over what exists

	Cloud assistants	Vendor wake-word SDKs	Hey Vayro
Where the audio goes	Vendor cloud	Depends on the product	Only to the owner's own server, only after a wake-up
Your own wake word	No	Licence, or large data collection plus a fee	Trained with our open pipeline
Hardware needed	Branded speaker	Often extra memory or audio chip	One low-cost chip, no extra memory
Licence / lock-in	Vendor account	Licence terms and fees	Open source (Apache-2.0)

Honest gap: today we wake by mistake about 16 times an hour with a TV and conversation, against about once an hour for commercial speakers in a published test. Closing that gap is our first engineering goal.

4 Real-world use cases

Private voice at home CORE WORKS

"Hey Vayro, turn on the fan."

The request goes to the family's own PC, not a vendor cloud. Detection, streaming and transcription work today; switching the fan is next.

A brand's own wake word NEXT

"Hey <Brand>" in an Indian-made appliance.

The same pipeline that trained "Hey Vayro" builds a new word with a measured test report. It needs multi-speaker data first.

Hands-free labs PILOT IDEA

A researcher in gloves logs a note by voice.

Audio stays on the campus server. This needs a lab-vocabulary recogniser and a real pilot. Assistive home use comes later, after testing with older voices.

5 Business and revenue model (open core)

The core firmware and model stay **open source**; that is how developers find and trust us. Revenue comes from what customers cannot easily do themselves:

Revenue stream	What the customer gets	Why it fits this project
1. Custom wake-word projects one-time fee per word	Their own wake word: voice collection, training, on-device tuning and a test report (accuracy, false wake-ups per hour, memory, CPU)	Our strongest skill is this repeatable train-verify-measure pipeline. FIRST REVENUE TARGET
2. Kits and deployments hardware + installation	Edge devices, a local speech server and set-up for a lab, office or home	Parts cost about Rs 2,000 per device plus an enclosure (Indian retail, Oct 2026)
3. Support and update plans annual	Model updates, over-the-air firmware updates, monitoring and maintenance	Installed devices need ongoing tuning. Annual maintenance contracts are familiar to Indian institutions
4. Integration services	Links to home or building automation, and Indian-language speech recognition on the server	Only the server changes; the device stays the same

Deliberately not chosen: a cloud service that processes customers' audio (it contradicts the product's purpose) and any data or advertising model. Prices will be set from pilot costs; we quote none yet.

Business Model Canvas · delivery	
Channels	Public GitHub and open-source communities; college incubator and startup cell; system integrators for institutions PLANNED
Relationships	Hands-on pilots, documented test reports, then support plans
Key partners	Potential, none signed yet: ESP32 module and PCB suppliers in India, a campus pilot site, open Indian-language speech models (AI4Bharat, Bhashini)

Business Model Canvas · engine	
Key resources	The working prototype, training and test tools, public code, report and 63-entry evidence log EXISTS
Key activities	Voice data collection, model training and testing, productization, pilots
Cost structure	Device parts, local servers, voice recording per new word, engineering time, certification for a custom board (later)

6 Productization roadmap

Stage	What we do	How we know it worked
1. Prototype DONE	Full chain built and measured on real hardware, Sep 2026	245.0 KiB RAM, 9.32% idle CPU, 138 ms median response, 20/25 for a new speaker
2. Robust product NEXT	5–10+ more voices and homes; TV-heavy training; a second check on the server; phone-based Wi-Fi set-up, over-the-air updates, encryption; real device control; power measured; small PCB and enclosure	Target: about 1 false wake-up per hour with TV while keeping at least 20/25 for every new speaker
3. Pilot PLANNED	One campus lab and a few homes for several weeks; a first custom wake word for one device maker	Targets met in daily use; written feedback from pilot users
4. Scale FUTURE	Custom wake-word service, kits and support plans; Indian-language recognition	Repeat customers; predictable cost and turnaround per new word

7 Why this can become a startup

- **Real and growing need:** 54% of US smart-speaker owners worry about the data collected (Pew, 2019), and in March 2025 Amazon removed Echo's local-processing option. Privacy-minded buyers have fewer choices.
- **Costly status quo:** a vendor custom wake word means a licence or 20,000+ recordings and a fee. We built ours from synthetic voices plus a small real set. It needs more voices to be robust, but the route is far lighter.
- **Cheap to make:** the whole system fits in 245.0 KiB without extra memory, so a low-cost module without it is enough (listed at Rs 469 each at volume).
- **Repeatable engine:** every new customer reuses the same training, verification and measurement tools.
- **Built for India:** Indian-language speech recognition can be added on the server; on-premise audio suits India's data-protection direction (DPDP Rules, main obligations from May 2027); parts are available locally.

SIH → beyond SIH. SIH is our first external validation, not the finish line. Whatever the result, we will keep going: more voices, fewer false wake-ups, a real pilot and a deployable product. The code, model and evidence stay open on GitHub.

Facts from our technical report and public repository: github.com/Kartikinfinity/vayro-edge-voice-activator. Commercial misactivation rate: Dubois et al., PoPETs 2020. Amazon change: TechCrunch, 15 Mar 2025. Custom wake-word requirements: Espressif ESP-SR documentation. Pew Research Center, 2019. Prices: Robu, Lion Circuits and Probots listings, Oct 2026. Items marked NEXT, PLANNED, PILOT IDEA or FUTURE are not built yet; no customers, partners or revenue exist today.